Helix OTA — Specification Corpus Index

Field	Value
Revision	1
Created	2026-06-07
Last modified	2026-06-07
Status	active
Status summary	Navigable master index of the entire Helix OTA specification corpus rooted at `docs/research/main_specs/`. Lists every document with a one-line purpose, a relative link, and a status (done / outline / future), grouped by corpus area (00-master foundation, research, additions, 1.0.0-MVP, 1.0.1, and the 1.X future-OS phases). Records how the corpus was produced and what was physically verified.
Status summary detail	This index is descriptive, not normative: the controlling rules live in [`00-master/documentation_standards.md`](00-master/documentation_standards.md). Where a referenced document is itself an outline or future-phase plan, that is reflected in its status column.
Issues	Exact HelixConstitution clause numbering (§7.1, §11.4.6, §11.4.29, §11.4.61, §11.4.123) is UNVERIFIED — carried from the corpus convention. There is no standalone 1.0.0-MVP `tests/` document yet; the four-layer testing strategy currently lives only in the master design (§testing strategy). The six NEW `ota-*` repositories are specified but not yet created (UNVERIFIED until they exist).
Issues summary	Index reflects files that physically exist as of Revision 1; planned-but-absent artifacts are marked `future`.
Fixed	N/A (initial revision).
Fixed summary	Initial corpus index.
Continuation	

Field	Value
	Add a row when the 1.0.0-MVP `tests/` spec is authored; promote the 1.X future-OS outlines to numbered phases; re-run the verification suite (see §8) in CI and link its output; reconcile UNVERIFIED clause numbers against the authoritative HelixConstitution.

Table of contents

> The Table-of-contents requirement is mandated by HelixConstitution §11.4.61 (UNVERIFIED). Every corpus document carries a metadata table followed by a ToC; this index does the same.

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1. Purpose

This README is the single entry point to the Helix OTA specification corpus. Each row below gives the document's one-line purpose, a relative link from this file, and a status. Use the status column to tell finished specifications apart from outlines and future-phase plans.

Per the anti-bluff convention (HelixConstitution §7.1 / §11.4.6, UNVERIFIED), nothing in this index is fabricated: every linked file physically exists in the corpus, every status reflects the linked document's own metadata, and any claim not confirmed from a real source is tagged `UNVERIFIED`. The full rule set lives in `00-master/documentation_standards.md` §8.

2. How this corpus was produced

The corpus was produced in three stages:

1. **Multi-agent research** — parallel research waves surveyed the OTA landscape and produced the twelve stack notes in `research/stacks/` plus the landscape

report, then reconciled the two operator input drafts in `additions/` into `research/additions_synthesis.md`.
2. **Adversarial review** — every contradiction (between the drafts, against the locked decisions, and against the HelixConstitution) was surfaced and resolved, and the evidence-backed choices were captured as the five ADRs in `research/adr/`. The master design and the threat model consolidate the surviving decisions, with the requirements traceability matrix acting as the completeness check that no operator requirement was dropped.
3. **Validation** — the executable/parseable 1.0.0-MVP artifacts (OpenAPI, SQL migrations, Kubernetes manifests, docker-compose) were run through real validators on the build host, and the export pipeline was exercised to render the corpus. The captured commands and outputs are recorded in `1.0.0-mvp/VALIDATION_EVIDENCE.md` and `00-master/export_pipeline.md`. See §8.

3. Status legend

Status	Meaning
done	Authored to full corpus depth; for executable artifacts, physically validated (see §8).
outline	Authored as a structured plan/outline; depth follows a completed phase.
future	Forward-looking research outline for a not-yet-scheduled phase.

4. 00-master — foundation

Foundation documents that anchor the whole corpus. Directory: `00-master/`.

Document	Purpose	Status
Master design specification	Canonical master design: vision, locked decisions, architecture, mandated stack, MVP scope, security/trust model, data model, rollout engine, telemetry, submodule map, phased layout, testing strategy, execution model.	done
Glossary	Canonical definitions of domain terms used across the corpus, each with where it appears.	done
Threat model	STRIDE-based threat model for the 1.0.0-MVP: twelve OTA threats with attack,	done

Document	Purpose	Status
	impact, MVP mitigation, residual risk.	
Submodule reuse map	Catalogue-first reuse/extend/new map binding each component to verified submodules; specifies the six NEW `ota-*` modules.	done
Documentation standards	Normative rules: mandatory metadata table, ToC, file naming, export targets, diagram conventions, cross-references, anti-bluff/UNVERIFIED, canonical submodule catalogue.	done
Requirements traceability matrix	Every operator requirement mapped to phase × component/spec × status (specified / researched / pending).	done
Export pipeline	The real, tested Markdown→HTML/DOCX(/PDF) + Mermaid→SVG/PNG pipeline (`scripts/export_docs.sh`) with captured live-run output.	done
Diagram renders (`_exports/`)	Generated artifacts: master-design diagram-01 in `.mmd`/`.svg`/`.png`, plus rendered `.html` and `.docx` of the master design.	done
ADR index	Architecture decisions ADR-0001..0005 live in `research/adr/` and are indexed in the master design §ADRs and in §5 below.	done

5. research — landscape, stack notes, ADRs, synthesis

Directory: `research/`.

Document	Purpose	Status
OTA landscape report	Landscape survey and engine-selection synthesis across the researched OTA stacks.	done
Additions synthesis	Reconciles the two operator drafts: extracts reusable elements, maps each to a destination, records every contradiction with a resolution rule.	done

Stack research notes — directory `research/stacks/` (12 notes):

Note	Purpose	Status
Android AVB / rollback	Android Verified Boot, dm-verity, `boot_control`, rollback.	done
Android `update_engine` API	`update_engine` API surface and OTA package structure.	done
Android 15 Virtual A/B	Android 15 Virtual A/B with compression — device-side core.	done
AOSP `update_engine`	AOSP `update_engine` / Android OTA mechanism.	done
Commercial / OSS fleet survey	balena, Toradex Torizon, Memfault, Foundries.io fleet platforms.	done
Eclipse hawkBit	hawkBit rollout management server.	done
Mender	Mender update client/ server.	done
OSTree	OSTree / libostree / rpm-ostree atomic file-based updates.	done
RAUC	RAUC fail-safe A/B bundle updater.	done
SWUpdate	SWUpdate embedded Linux `.swu` updater.	done
TUF + go-tuf/v2	TUF supply-chain integrity with go-tuf/v2.	done
Uptane	Uptane automotive-grade trust framework.	done

Architecture Decision Records — directory `research/adr/` (5 ADRs, status Proposed):

ADR	Purpose	Status
ADR-0001 Wrapped engine	hawkBit vs Mender vs AOSP-native-only.	done (Proposed)
ADR-0002 Supply-chain trust	Plain signing vs TUF vs Uptane, and MVP timing.	done (Proposed)
ADR-0003 Server topology	Modular monolith vs microservices, with scale trigger.	done (Proposed)
ADR-0004 Transport	HTTP/3 (QUIC) + Brotli with HTTP/2 fallback.	done (Proposed)
ADR-0005 Delta updates	AOSP block diffs vs custom, and phase placement.	done (Proposed)

6. additions — operator inputs

Raw operator-supplied input drafts, preserved as the source for the synthesis (§5). Directory: `additions/`.

Document	Purpose	Status
Initial research (draft 1)	First operator draft of the Helix OTA plan and research.	done (operator input)
Initial research (draft 2)	Second operator draft (parallel-wave research framing).	done (operator input)

7. Phase specifications

7.1 1.0.0-mvp

The first shippable phase: native Android 15 A/B on device, a Go + Gin control plane, a React dashboard, and a PostgreSQL + MinIO/S3 artifact store. Directory: `1.0.0-mvp/`.

API — `1.0.0-mvp/api/`:

Document	Purpose	Status
Endpoint specification	The `/api/v1` REST endpoint specification.	done
OpenAPI 3.1 description	Machine-readable OpenAPI 3.1 (12 paths, 24 schemas); lint-validated (§8).	done (validated)

Database — `1.0.0-mvp/database/`:

Document	Purpose	Status
Schema	The 1.0.0-MVP PostgreSQL schema.	done
Migrations (up/down)	`001_initial_schema` up + down SQL; applied against a live Postgres (§8).	done (validated)

Security — `1.0.0-mvp/security/`:

Document	Purpose	Status
Key management	MVP signing-key lifecycle and custody.	done
Signing & verification	Artifact signing and verification flow.	done
Transport security	TLS/transport security for the control plane and clients.	done

Server — `1.0.0-mvp/server/`:

Document	Purpose	Status
Architecture	Modular-monolith server architecture.	done
Artifact validation	Artifact upload validation pipeline.	done
Telemetry processing	Telemetry processing and device health.	done

Android client — `1.0.0-mvp/client_android/`:

Document	Purpose	Status
Integration guide	Android agent integration guide.	done
Build integration	Android agent build integration.	done
`update_engine` integration	Apply path via AOSP `update_engine`.	done
Code snippets	Reference Kotlin/AOSP build snippets (downloader, verifier, applier, poll worker, telemetry, `Android.bp`, etc.).	done (snippets; compile pending toolchain)

Deployment — `1.0.0-mvp/deployment/`:

Document	Purpose	Status
Overview	Deployment overview for the MVP.	done
MinIO setup	MinIO/S3 artifact-store setup.	done
docker-compose	Single-host MVP compose stack.	done
Kubernetes manifests	namespace, ota-server deployment + service, postgres statefulset; kubeconform-validated (§8).	done (validated)

Cross-cutting:

Document	Purpose	Status
VALIDATION_EVIDENCE	Physical, reproducible validation of the MVP artifacts with real tools — commands + captured output.	done
Tests	Four-layer testing strategy is currently only in the master design; no standalone `tests/` spec exists yet (UNVERIFIED).	future

7.2 1.0.1-staged-rollout

Document	Purpose	Status
Phase 1.0.1 plan	First post-MVP phase: staged rollout, health-gated halt, device-side trust (TUF), end-user rollback.	outline

7.3 1.X future-OS phases

Each extends Helix OTA to a new OS family via the OS-adapter seam; the control plane is reused unchanged and only a device adapter is added.

Document	Purpose	Status
1.X Linux	Linux support (RAUC / OSTree-bootc / SWUpdate / package-manager backends).	future
1.X Windows		future

Document	Purpose	Status
	Windows 10/11, IoT/ LTSC, Server support.	
1.X Other OS	macOS, *BSD, RTOS, and other upstream OSes.	future

8. Verification evidence

> The following were physically executed; do not treat as claims. Full commands and captured output are in `1.0.0-mvp/VALIDATION_EVIDENCE.md` and `00-master/export_pipeline.md`.
>
> - **Live PostgreSQL migration apply** — both `001_initial_schema` up + down migrations applied against a live server with `psql -v ON_ERROR_STOP=1` (verified).
> - **OpenAPI lint** — `npx @redocly/cli lint api/openapi.yaml` → valid (1 cosmetic `info-license` warning); 12 paths, 24 component schemas (verified).
> - **kubeconform** — `kubeconform -summary -strict deployment/kubernetes/*.yaml` over the four manifests (verified).
> - **Export render** — `scripts/export_docs.sh` rendered Markdown→HTML/DOCX and Mermaid→SVG/PNG (`pandoc` + `mmdc` present); PDF requires a LaTeX engine that was absent on the build host, so PDF is deferred to a CI container (UNVERIFIED) rather than faked.
>
> Validations requiring tools absent on the build host (`docker`, `yq`, LaTeX, `drawio`, `plantuml`, `dot`) are deferred to a CI container, not faked (UNVERIFIED until that job runs).

9. Submodule reference (catalogue + the six NEW ota-* modules)

This index references only real catalogue submodules and the six NEW `ota-*` modules. The canonical catalogue names are fixed in `00-master/documentation_standards.md` §9; the bindings (reuse / extend / new) are in `00-master/submodule_reuse_map.md`.

The six NEW modules to be created (specified, not yet created — UNVERIFIED until they exist):

NEW module	Purpose
`ota-protocol`	Shared wire types, manifest schema, status/event enums (Go + KMP); pure contracts.
`ota-artifact-validator`	Structure / hash / signature / metadata validation pipeline; no transport.
`ota-rollout-engine`	Staged-rollout + halt/advance logic, OS-agnostic; no HTTP.

NEW module	Purpose
`ota-update-engine-bridge`	Android-only wrapper over AOSP `update_engine` / `boot_control`.
`ota-android-agent`	KMP device agent: poll, download, verify, apply, report.
`ota-telemetry-schema`	Telemetry event / metric schema + codecs; shared server + agent.